Thesis progress

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DEGLI STUDI
DI MILANO

Index

1 1 ° week

GitHub

The whole code is available at this GitHib link

1 ° week

Uploading file .root

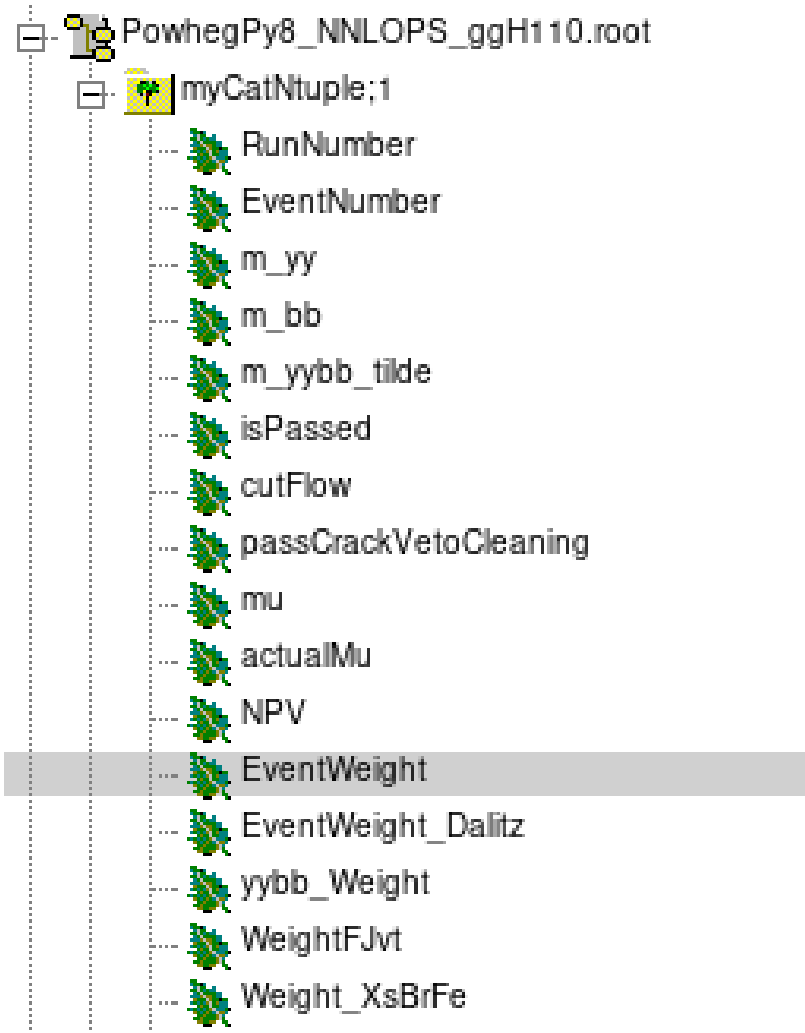
The file uploaded are:

- PowhegPy8_NNLOPS_ggH110.root;
- PowhegPy8_NNLOPS_ggH125.root;
- PowhegPy8_NNLOPS_ggH130.root;
- PowhegPy8_NNLOPS_ggH140.root;

Data

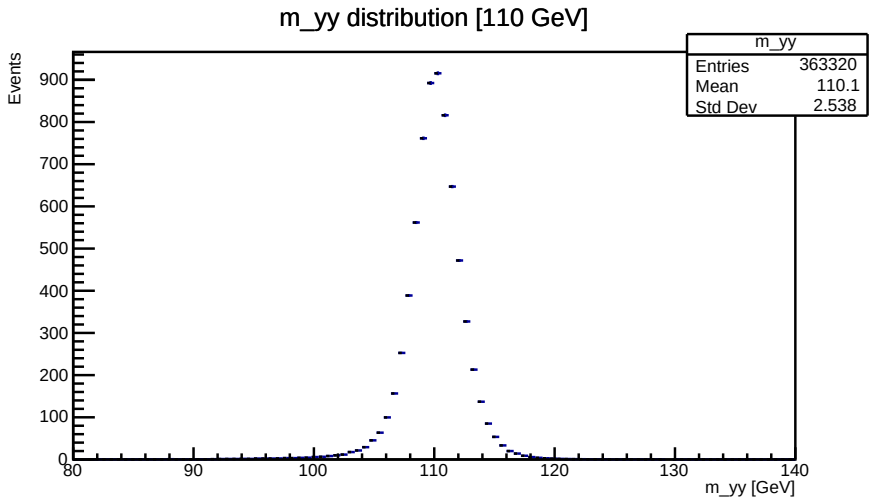
In each file, there is a `TTree myCatNtuple`, which contains all the variables. Only three of them were used:

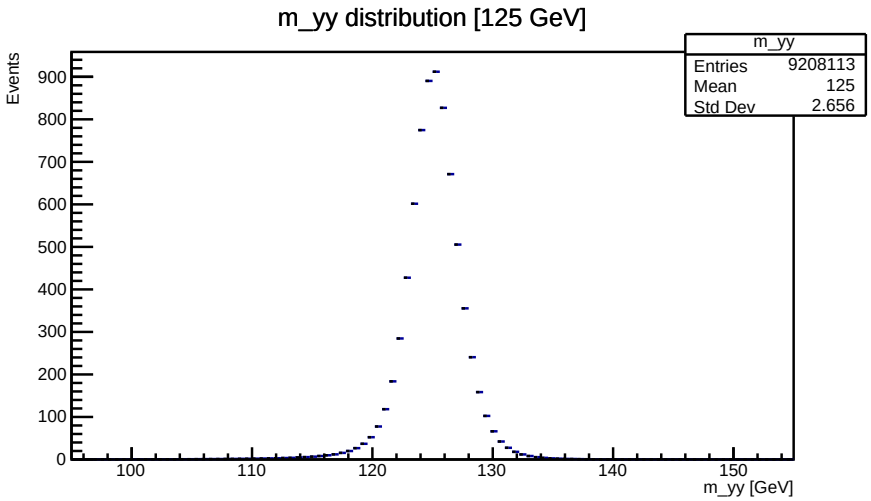
- **m_yy**: mass distribution;
- **EventWeight**: it contains all type of weights.
- **cutFlow**

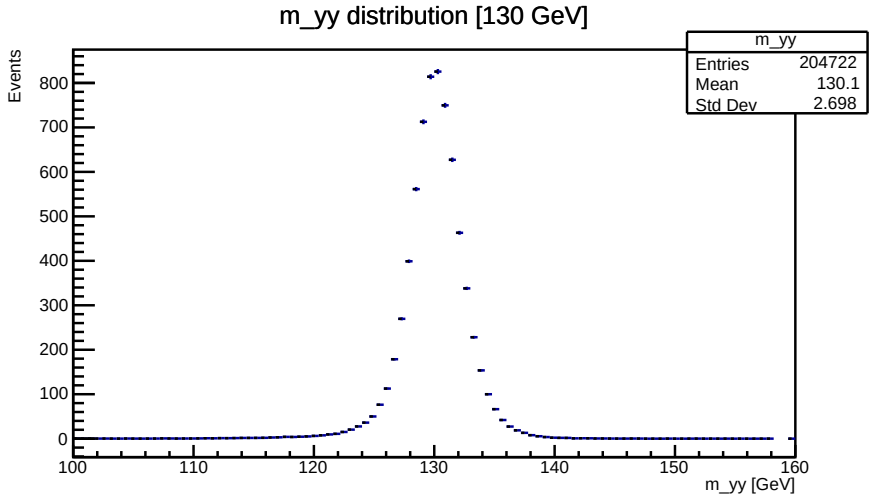


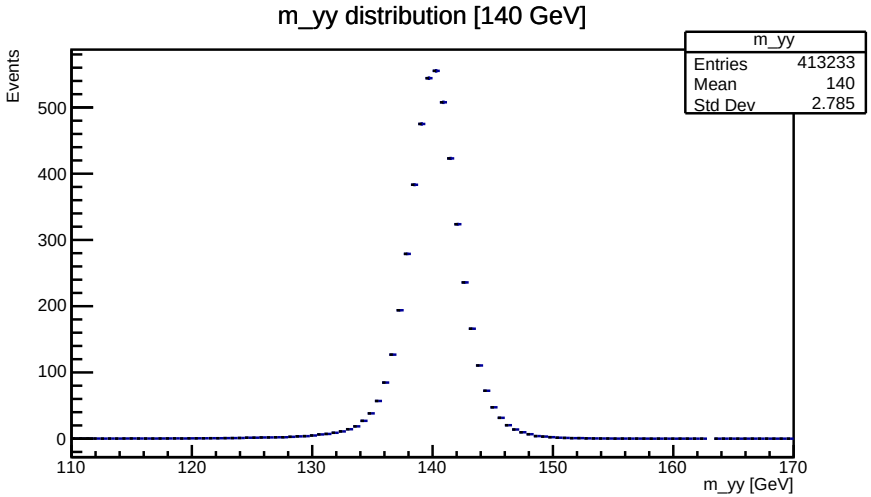
$m_{\gamma\gamma}$ distibution

Using all entries the distributions at different masses are:









Data analysis and fits

The distributions are fitted using two type of fit:

- **gaussian fit**: there are 2 parameters:
  - mean μ;
  - sigma σ;

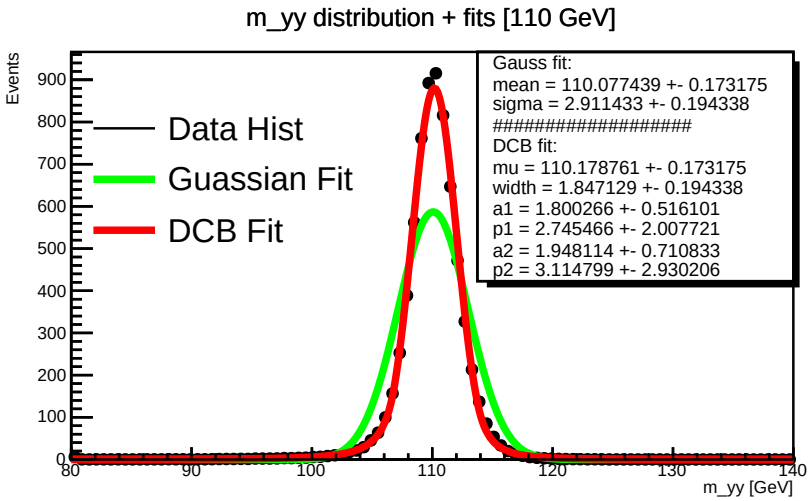
$$f(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

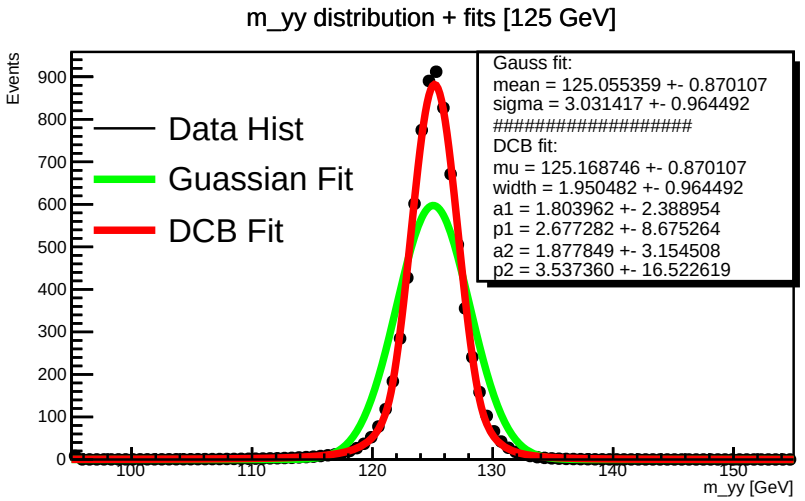
- **DCB fit**: there are six parameters:
  - mean μ;
  - sigma σ
  - a_1;
  - p_1;
  - a_2;
  - p_2;

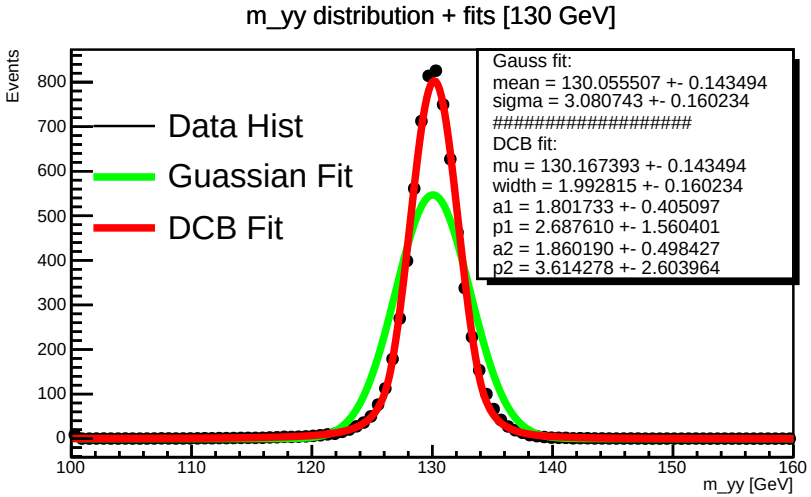
$$f(x, \mu, \sigma, a_1, p_1, a_2, p_2) = \cdots$$

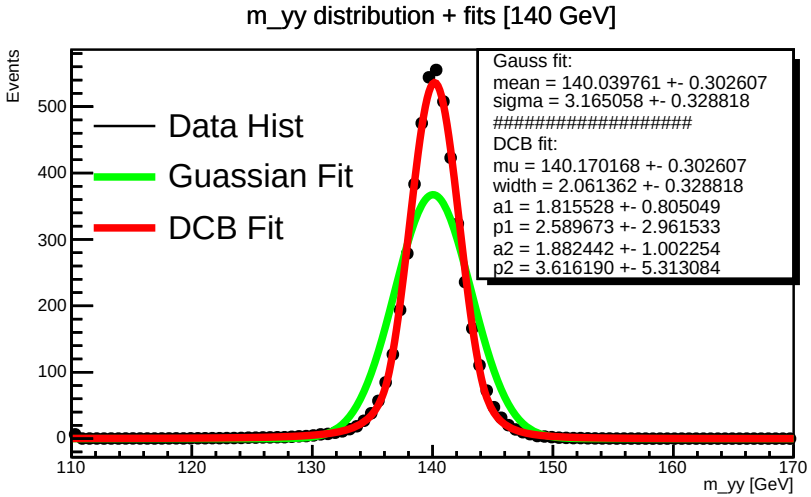
Gaussian and DCB fits

Applying the fits to all entries:









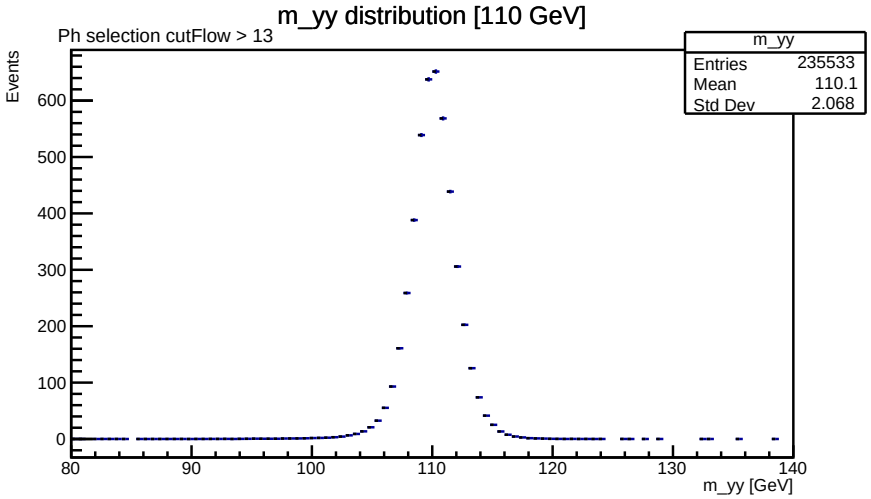
CutFlow

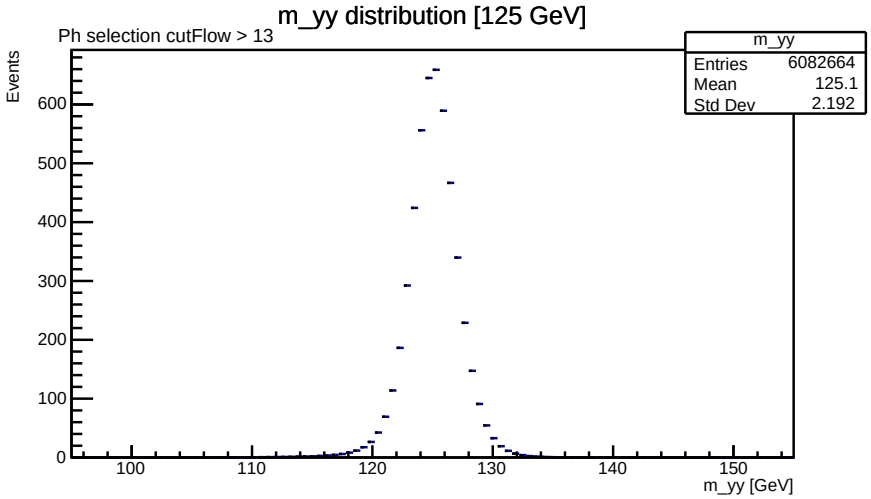
Events with at least two isolated γ with tight identification could be selected, where the transverse momentum of the leading photon is such that $p_T/m_{\gamma\gamma} > 0.35$, whereas that of the sub-leading photon is $p_T/m_{\gamma\gamma} > 0.25$. Then there are also other demands on the "quality" of data collection and triggers.

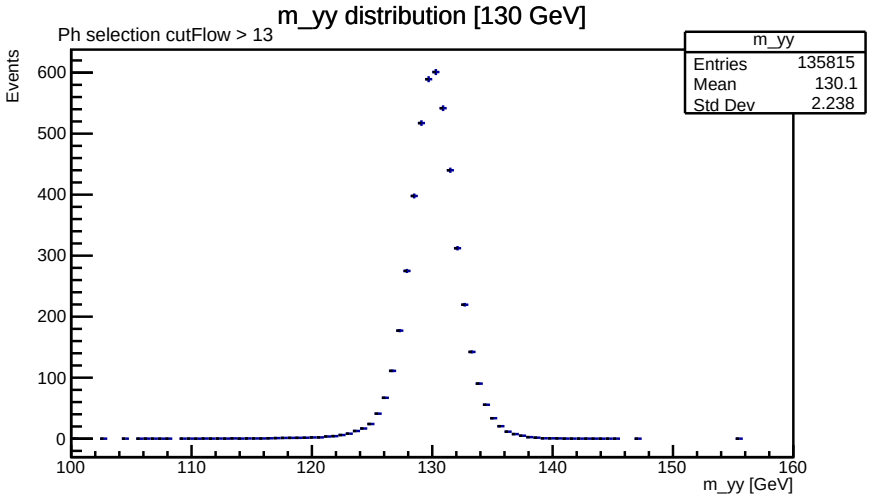
$\Longrightarrow$ Using the `cutFlow` greater than 13, the events are selected.

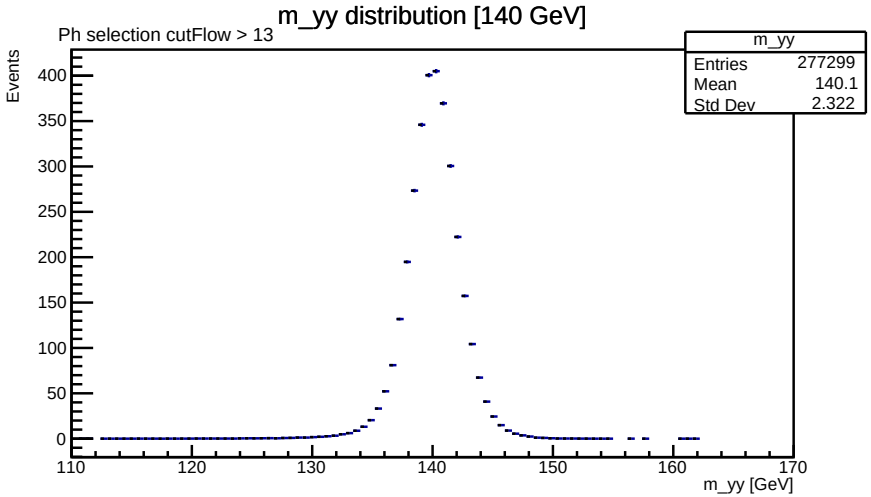
$m_{\gamma\gamma}$ distribution with cutFlow > 13

Using `cutFlow`>13 entries the distributions at different masses are:



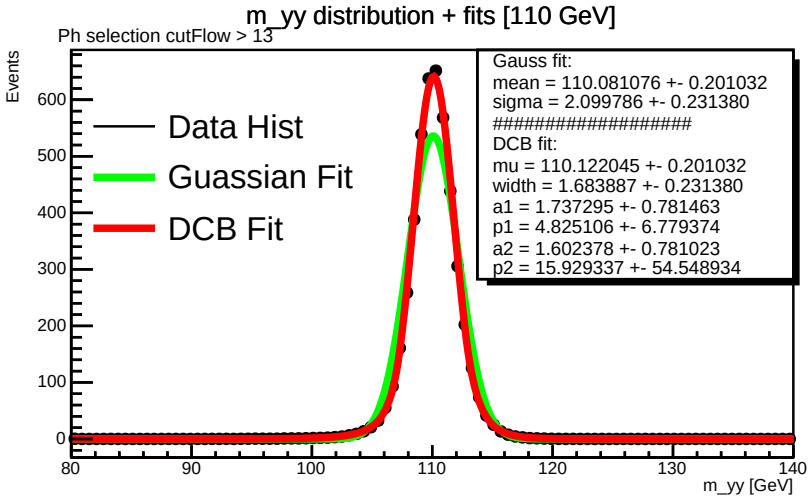


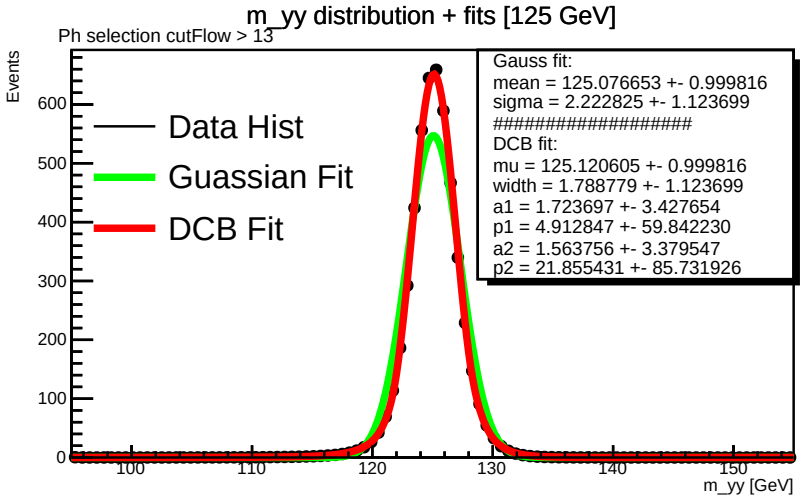


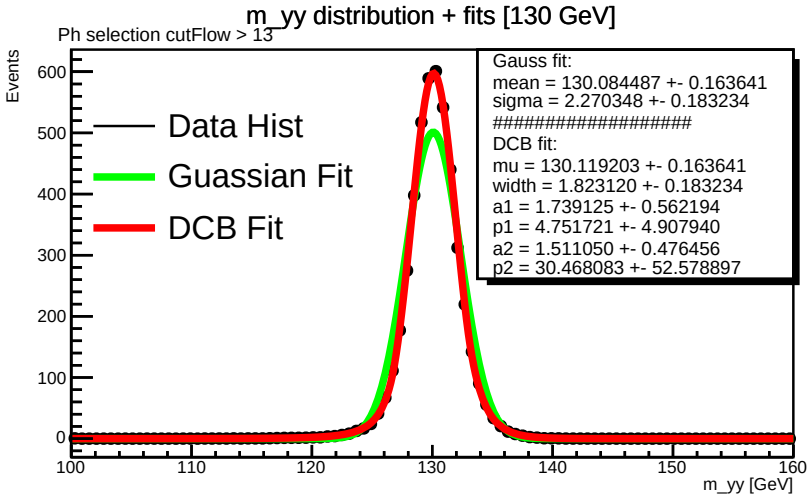


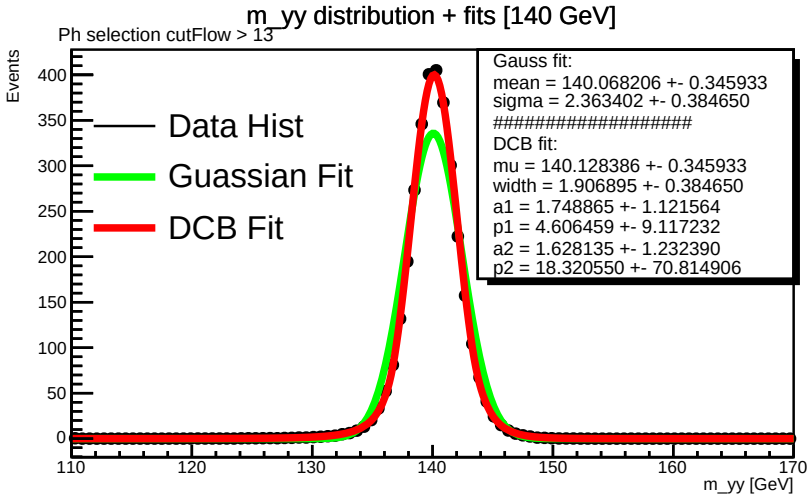
Gaussian and DCB fits with cutFlow > 13

Applying the fits to all entries:



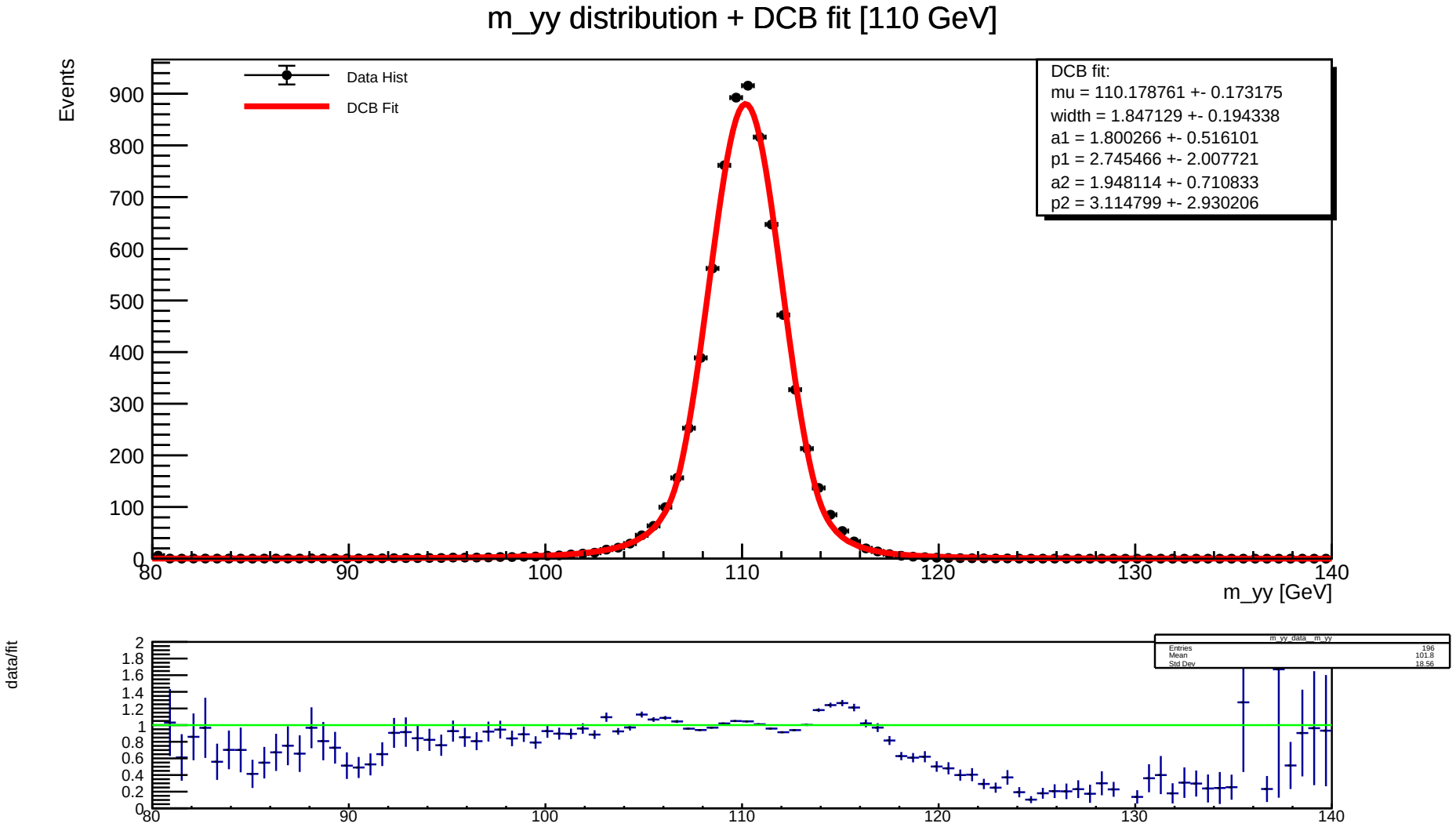






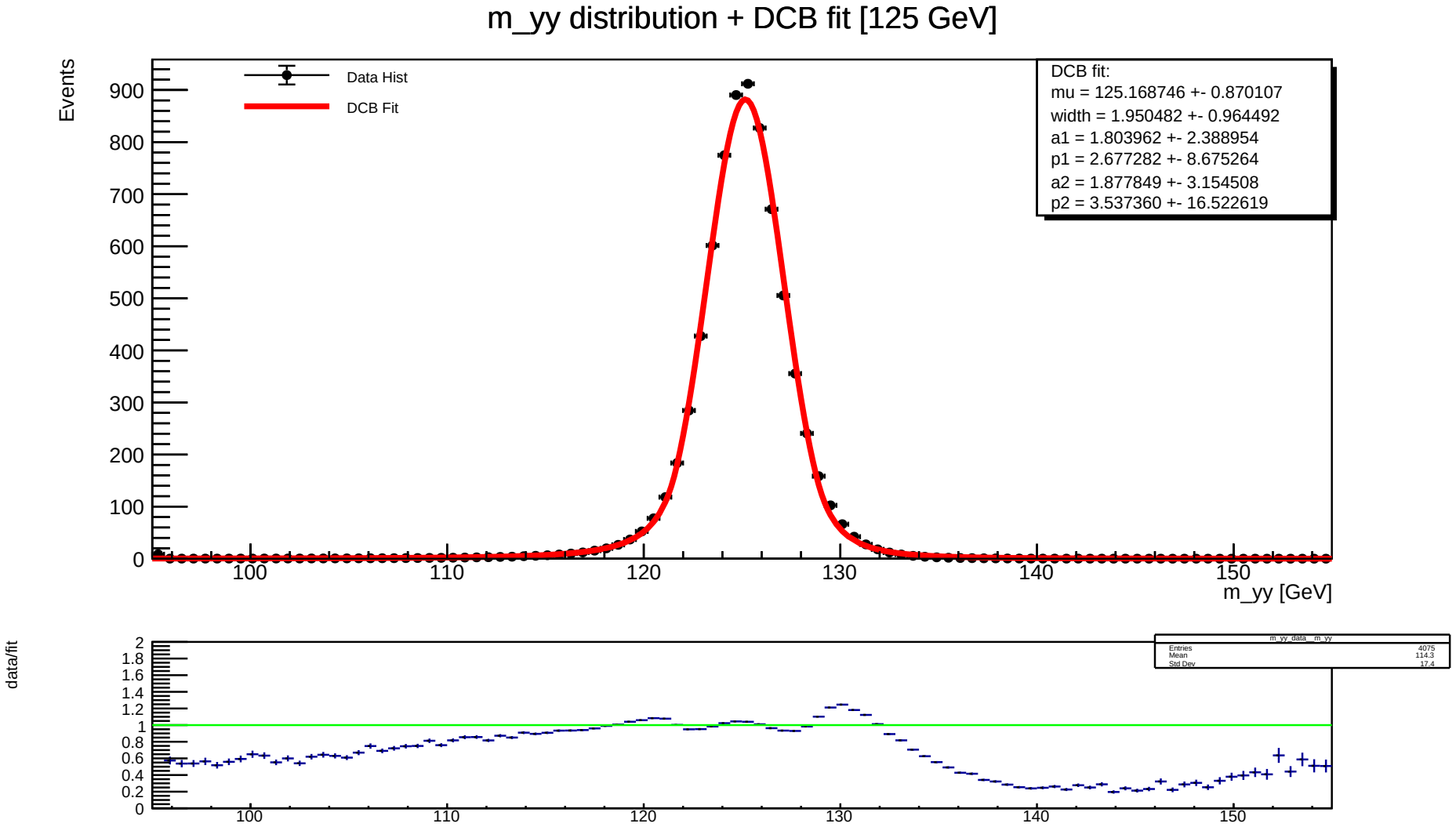
DCB fit and ratio plot [110 GeV]

The `ratio plot` between the fit and the distribution is added to each DCB fit. Using the 110 GeV mass distribution:



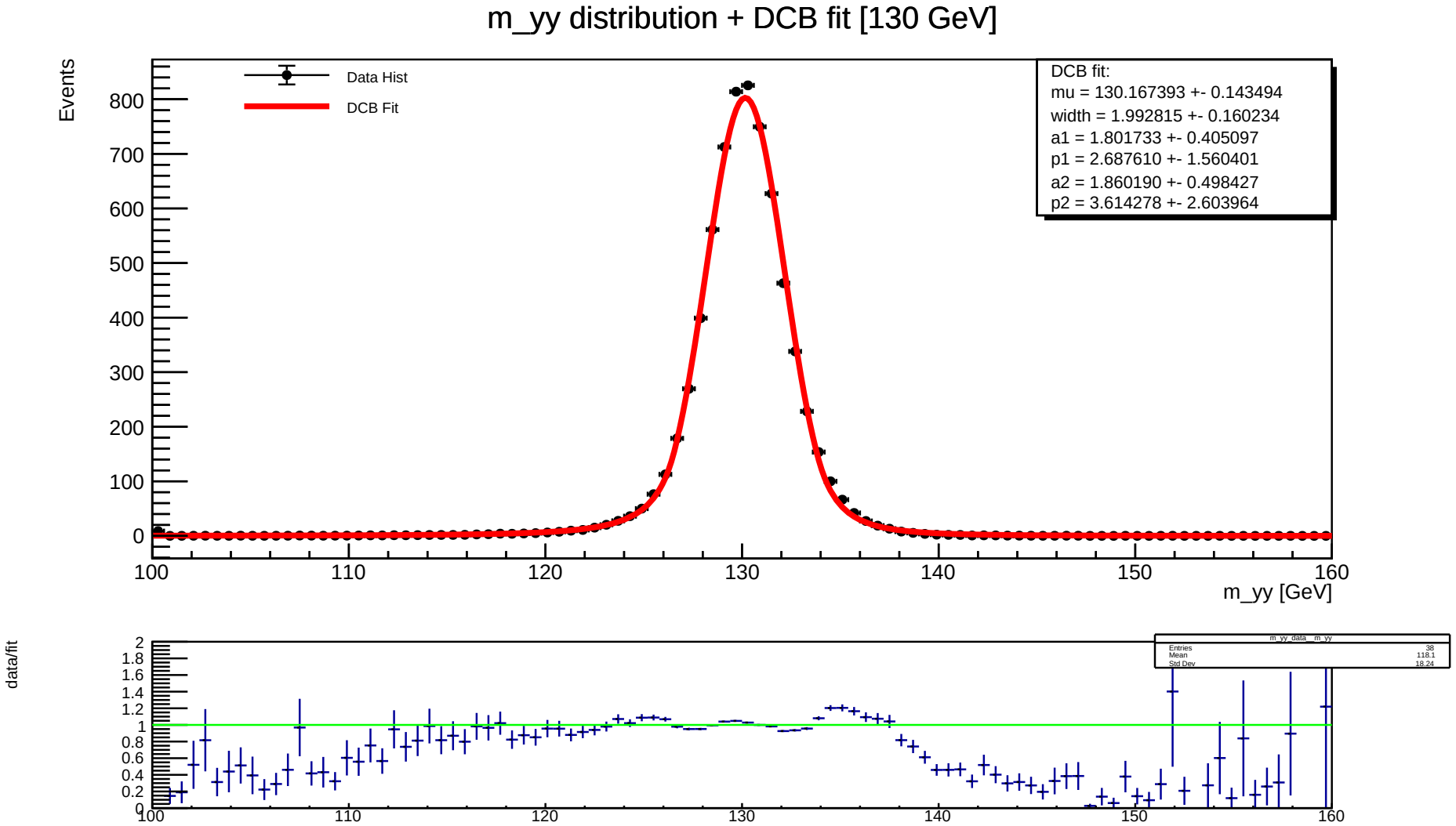
DCB fit and ratio plot [125 GeV]

The `ratio plot` between the fit and the distribution is added to each DCB fit. Using the 125 GeV mass distribution:



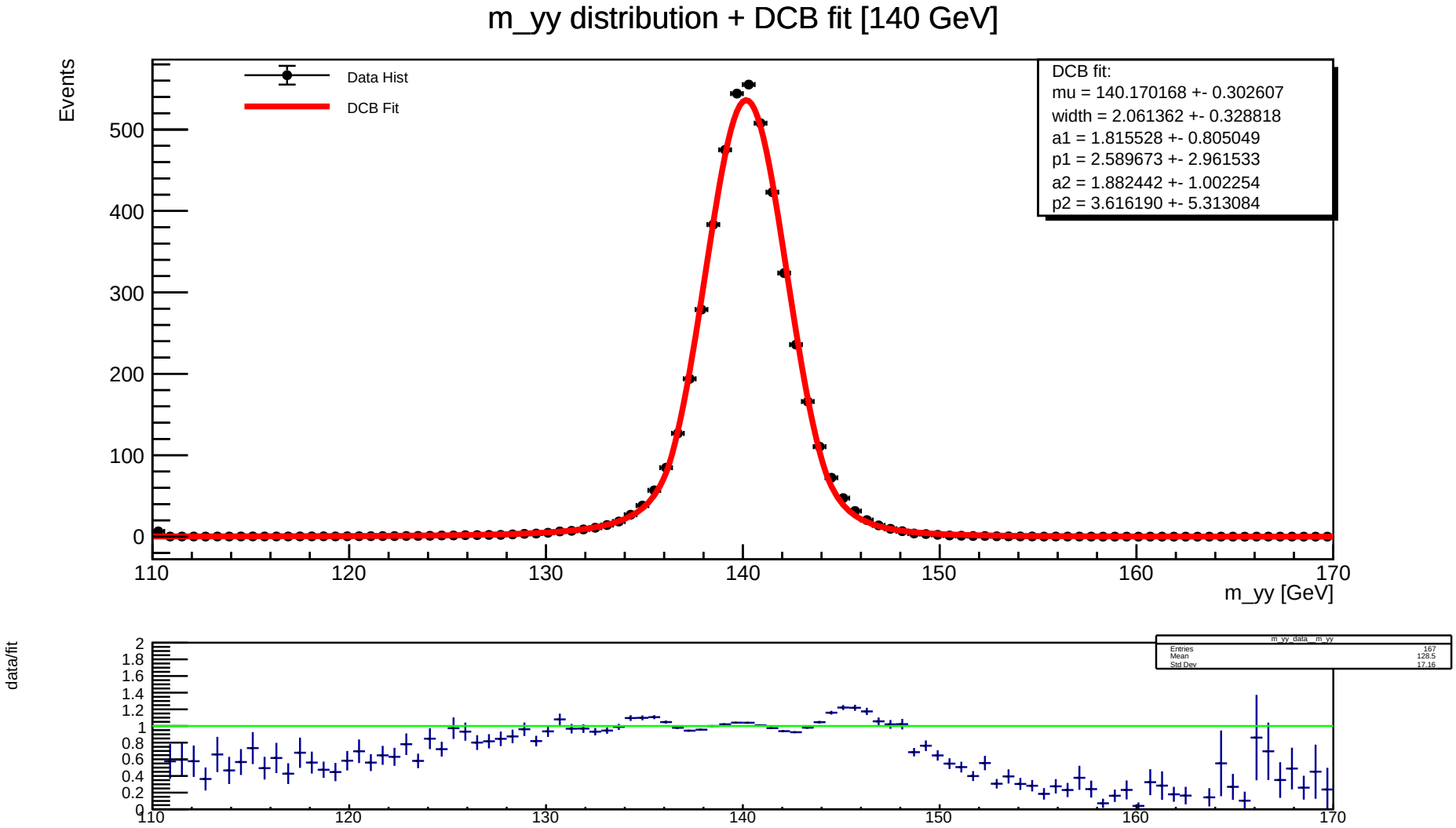
DCB fit and ratio plot [130 GeV]

The `ratio plot` between the fit and the distribution is added to each DCB fit. Using the 130 GeV mass distribution:



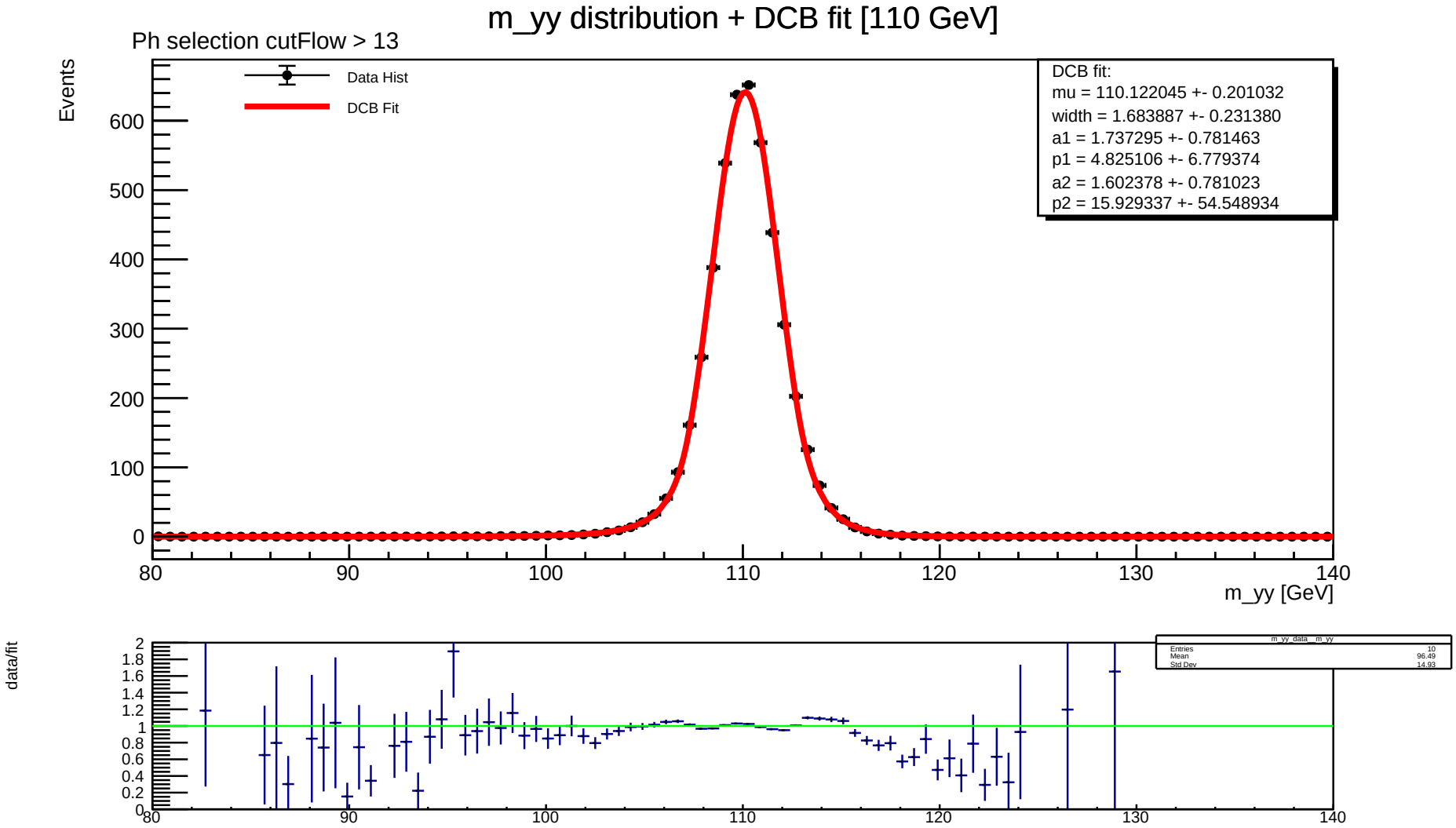
DCB fit and ratio plot [140 GeV]

The `ratio plot` between the fit and the distribution is added to each DCB fit. Using the 140 GeV mass distribution:



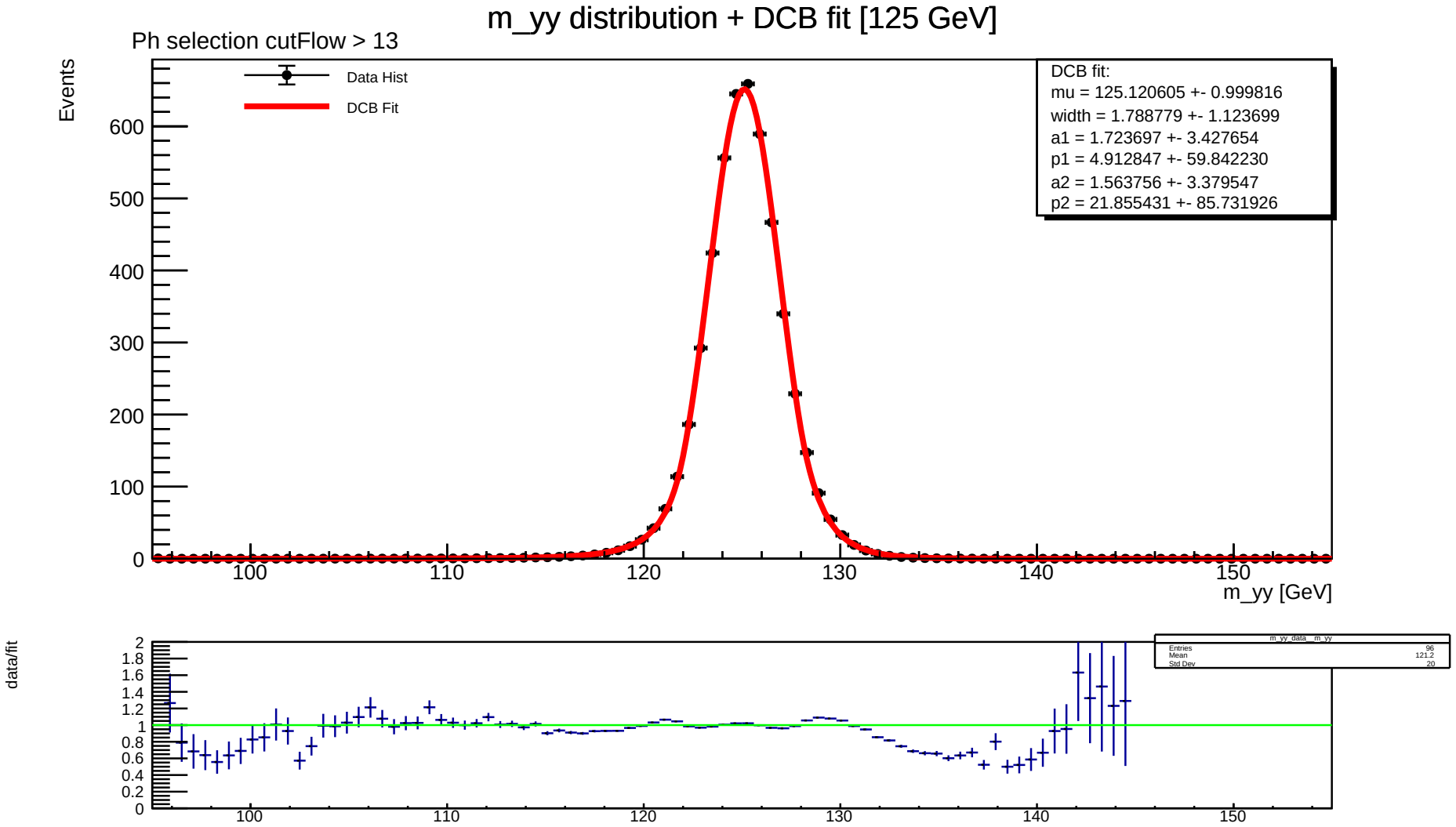
DCB fit+ratioplot [110 GeV] (cutFlow > 13)

The `ratio plot` between the fit and the distribution is added to each DCB fit. Using the 110 GeV mass distribution with `cutFlow` > 13:



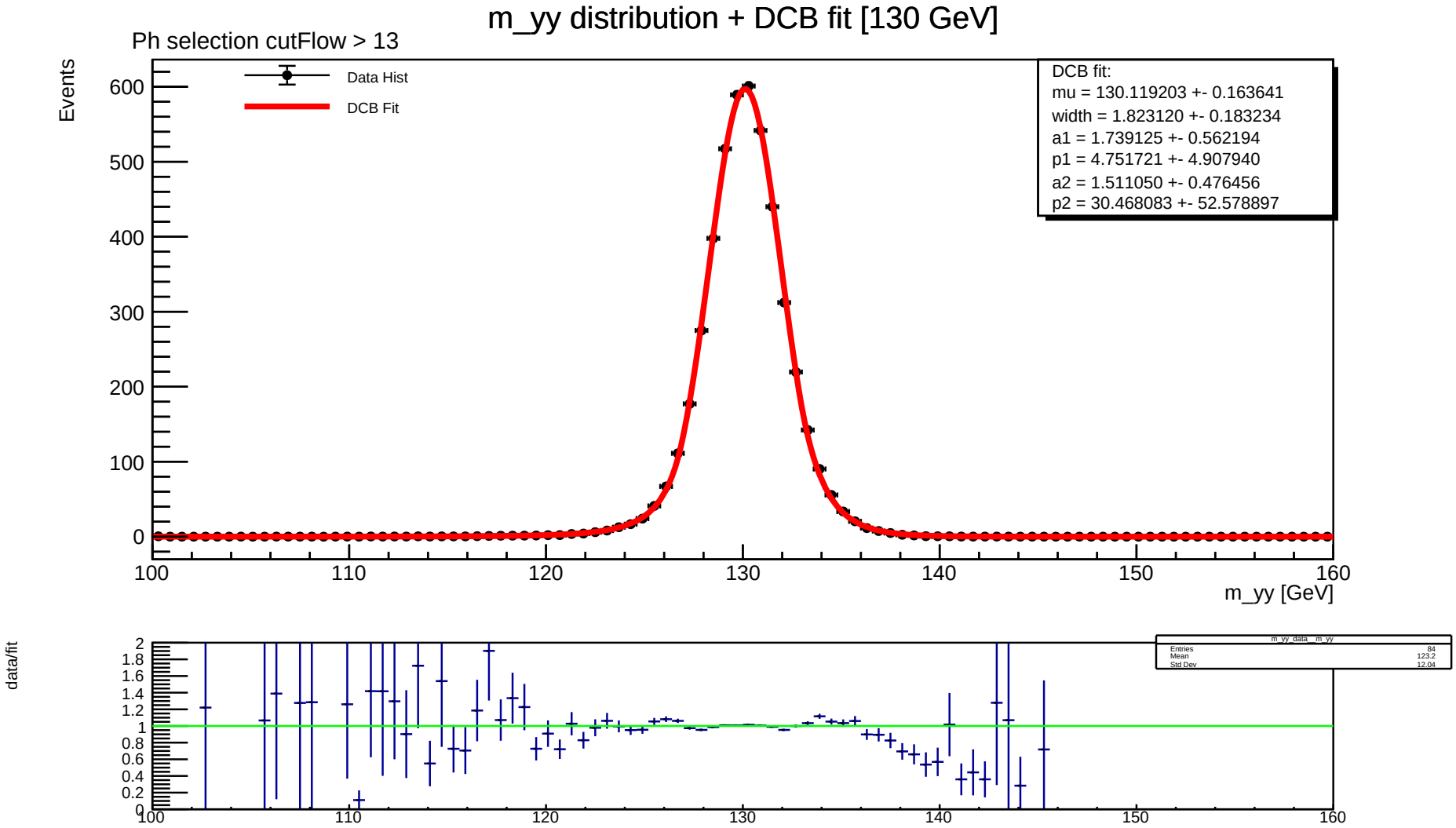
DCB fit+ratioplot [125 GeV] (cutFlow > 13)

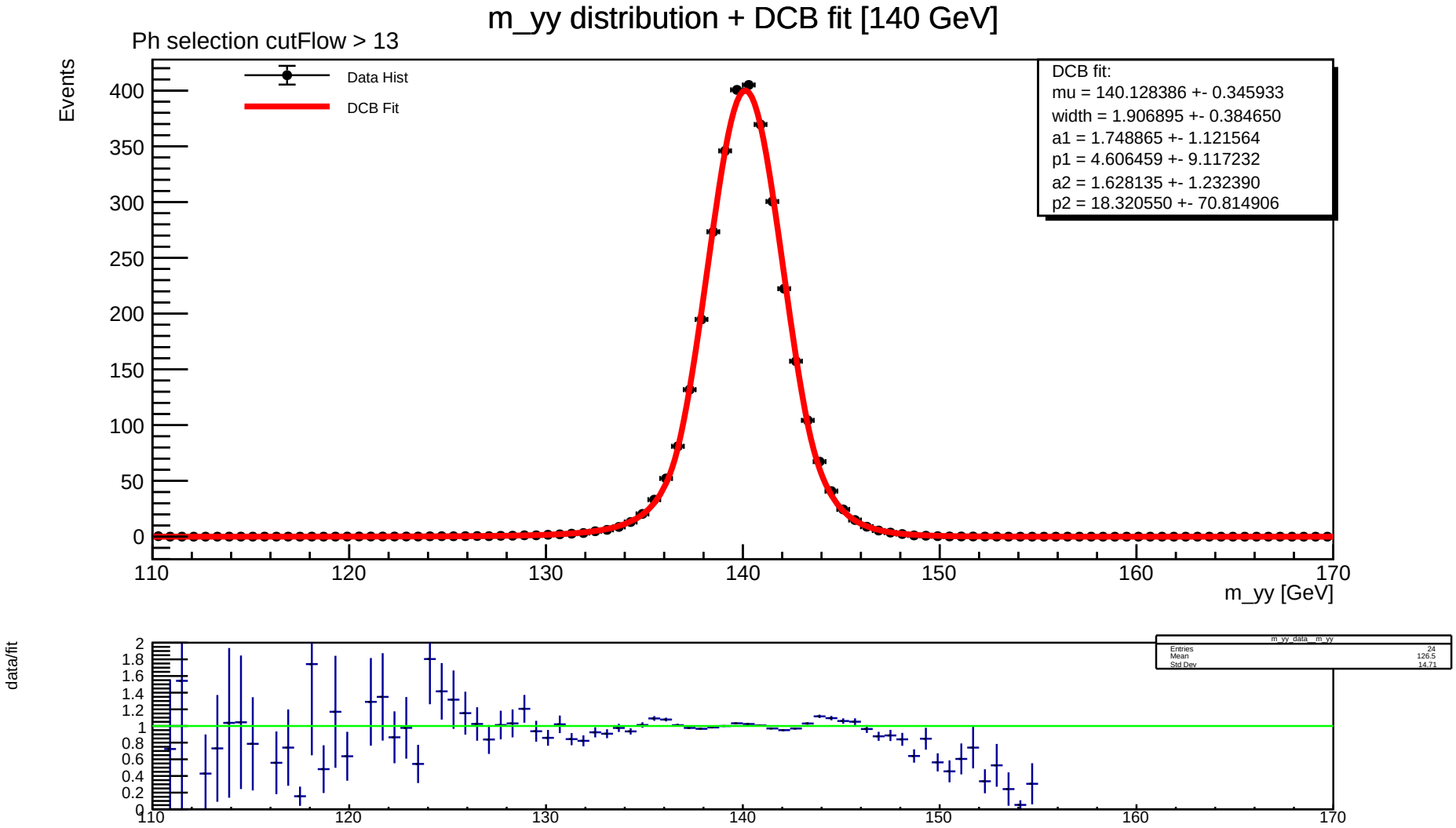
The `ratio plot` between the fit and the distribution is added to each DCB fit. Using the 125 GeV mass distribution with `cutFlow` > 13:



DCB fit+ratioplot [130 GeV] (cutFlow > 13)

The `ratio plot` between the fit and the distribution is added to each DCB fit. Using the 130 GeV mass distribution with `cutFlow` > 13:



DCB fit+ratioplot [140 GeV] (cutFlow > 13)

The `ratio plot` between the fit and the distribution is added to each DCB fit. Using the 140 GeV mass distribution with `cutFlow` > 13: